

Assignment-2

→ Microservices & Event-Driven Architecture in E-Commerce

Scenario: Building a Scalable online Marketplace.

• Architecture

• Microservices:

- Services for Inventory, Orders, Payments, Notifications, and user Management.

- Each service has its own database and communicates over HTTP/gRPC

• Event-Driven Architecture:

- uses a message broker (e.g., Kafka) for async communication

- Events: Order Placed, Payment Processed, Item Shipped.

• SOLID Principles Applied

- S: Single Responsibility - Each Microservice handles one domain.

- O: Open/Closed - Services can be extended via events without modifying existing code.

- L: Liskov Substitution - Interfaces used for shared functionality (e.g., payment gateways).

- I: Interface Segregation - Services expose only relevant APIs.

- D: Dependency Inversion - Services depend on abstraction (e.g., event interfaces).

DRY and KISS Principles

• DRY:

- Centralized Utility Services (e.g., logging, auth)
- Reusable event Schema formats.

• KISS:

- Simple API Contracts.
- Decentralized business logic to avoid monolith complexity.